Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE

CS2311: SOFTWARE ENGINEERING

Division	SY CS-A
Batch	3
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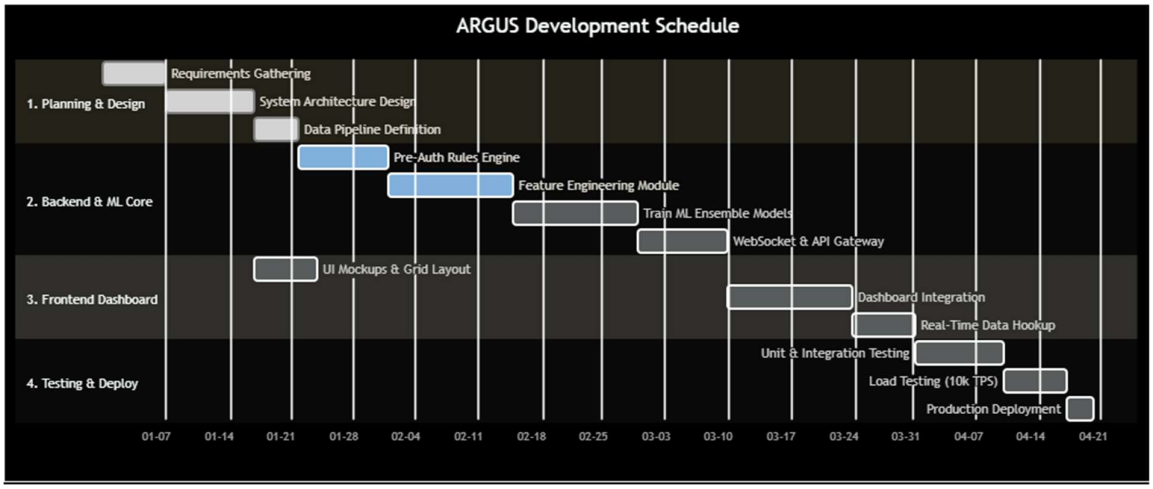
ASSIGNMENT 7

PROBLEM STATEMENT

- Prepare a software project plan with scheduling using Gantt charts and PERT charts for a sample project.

GANTT CHART



Gantt Summary Table

Task	Start	End	Duration	Dependencies
SoW Writing	W1	W2	2 weeks	–
Architecture Design	W2	W3	2 weeks	SoW
DFD / UML Diagrams	W3	W4	2 weeks	Architecture
DB Schema Design	W3	W4	1 week	Architecture
FastAPI Setup	W4	W5	1 week	DB Schema
REST API + WebSocket	W5	W6	2 weeks	FastAPI Setup
Data Prep & EDA	W4	W5	1 week	DB Schema
Feature Engineering	W5	W7	3 weeks	Data Prep
XGBoost Training	W6	W7	2 weeks	Feature Eng.
Isolation Forest	W6	W7	2 weeks	Feature Eng.
Ensemble + Threshold	W7	W8	1 week	XGBoost + IF
Pre-Auth Engine	W5	W7	2 weeks	REST API
Device Intelligence	W6	W7	2 weeks	Pre-Auth
Graph Fraud Detector	W7	W8	2 weeks	Pre-Auth
Phishing Protection	W7	W8	2 weeks	Pre-Auth
Explainable AI	W8	W9	2 weeks	All ML modules

Alert & Case Mgmt	W8	W9	2 weeks	REST API
React/Vite Setup	W5	W6	1 week	FastAPI
Dashboard UI	W6	W8	3 weeks	React Setup
Charts & Simulation	W8	W9	2 weeks	Dashboard UI
Integration Testing	W9	W10	2 weeks	All above
Performance Tuning	W9	W10	2 weeks	Integration
Bug Fixes	W10	W11	1 week	Testing
README / SoW Final	W10	W11	1 week	Testing
Pitch & Demo Prep	W11	W12	2 weeks	All complete

PERT CHART

PERT Formula

```
O = Optimistic time (best case)
M = Most Likely time
P = Pessimistic time (worst case)
Expected Time: tₑ = (O + 4M + P) / 6
Variance: σ² = ((P − O) / 6)²
```

Task Estimates

Task ID	Task Name	O	M	P	t_e	σ^2
T1	SoW & Requirements	1	2	3	2	0.11
T2	System Architecture	1	2	4	2.2	0.25
T3	DFD + UML Diagrams	1	1.5	3	1.7	0.11
T4	DB Schema	0.5	1	2	1.1	0.06
T5	FastAPI Backend Setup	0.5	1	2	1.1	0.06
T6	REST API + WebSocket	1	2	4	2.2	0.25
T7	Data Prep & EDA	1	1.5	3	1.7	0.11
T8	Feature Engineering	2	3	5	3.2	0.25
T9	XGBoost Training	1	2	4	2.2	0.25
T10	Isolation Forest	0.5	1	2	1.1	0.06
T11	Ensemble Model	0.5	1	2	1.1	0.06
T12	Pre-Auth Engine	1	2	3	2	0.11
T13	Device Intelligence	1	2	4	2.2	0.25
T14	Graph Fraud Detector	1	2	4	2.2	0.25
T15	Phishing Protection	1	1.5	3	1.7	0.11
T16	Explainable AI	1	2	4	2.2	0.25
T17	Alert & Case Mgmt	1	2	3	2	0.11
T18	React Dashboard	1	2	4	2.2	0.25
T19	Charts & Simulation	1	2	3	2	0.11
T20	Integration Testing	1	2	4	2.2	0.25
T21	Performance Tuning	0.5	1	2	1.1	0.06
T22	Documentation	0.5	1	2	1.1	0.06
T23	Pitch & Demo Prep	0.5	1	2	1.1	0.06

PERT Network Diagram

```
(START)
   |
   T1: SoW (2.0w)
   |
   T2: Architecture (2.2w)
   |         \
   T3:DFD    T4: DB Schema (1.1w)
   (1.7w)    |           \
             T5:FastAPI   T7:Data Prep
             (1.1w)       (1.7w)
             |             |
             T6:API(2.2w)  T8:Feature Eng.(3.2w)
             |             |       \
             |          T9:XGB   T10:IF
             |          (2.2w)   (1.1w)
             |             \      /
             |           T11:Ensemble(1.1w)
             |                 |
         T12:Pre-Auth(2.0w)   T16:XAI(2.2w)
         /   \                     |
      T13   T14                    |
      T15   T17                    |
       \     |    T18:React(2.2w)  |
        \    |    T19:Charts(2.0w) |
         \   |         |           |
         T20: Integration Testing (2.2w)
              |
         T21: Perf Tuning (1.1w)
              |
         T22: Documentation (1.1w)
              |
         T23: Demo Prep (1.1w)
              |
           (END)
```

Critical Path Calculation

Task	t_e (weeks)	σ^2
T1: SoW	2.0	0.11
T2: Architecture	2.2	0.25
T4: DB Schema	1.1	0.06
T8: Feature Engineering	3.2	0.25
T9: XGBoost Training	2.2	0.25
T11: Ensemble	1.1	0.06
T12: Pre-Auth Engine	2.0	0.11
T16: Explainable AI	2.2	0.25

T20: Integration Testing	2.2	0.25
T21: Performance Tuning	1.1	0.06
T22: Documentation	1.1	0.06
T23: Demo Prep	1.1	0.06
TOTAL	**21.5 weeks**	**1.77**

Resultant PERT Chart

Milestones

Milestone	Target Week	Description
M1: Design Complete	End of W4	SoW, architecture, DFDs, DB schema finalized
M2: Backend Live	End of W6	FastAPI running with REST API and WebSocket
M3: ML Models Trained	End of W8	XGBoost + ensemble + all advanced modules working
M4: Frontend Complete	End of W9	Full React dashboard with live simulation
M5: System Integrated	End of W10	Full end-to-end system verified
M6: Demo Ready	End of W12	Pitch, demo video, documentation complete